

Smaller and Shallower XOR Circuits for AES MixColumns: 97 Gates at Depth 3; 92 at Depth 4; 89 at Depth 5; and an Independent 88 at Depth 7

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Abstract

We give four verified implementations of AES MixColumns as circuits of 2-input XOR gates over $\text{GF}(2)$. Three improve the published depth-count Pareto frontier at their depth: **97** gates at depth **3**, the minimum depth, against 99 [2]; **92** at depth **4** against 97 [3]; **89** at depth **5** against 94 [3]. The fourth, **88** at depth **7**, *ties* the published gate-count floor of Jean [9] — who has priority — with an independent circuit sharing 61 of 88 internal masks. All were found by stochastic local search over a value-set formulation of the shortest-linear-program problem. No 87 was found: 47 canonical 88-gate circuits have exhaustively empty remove- ≤ 3 neighbourhoods. Nothing here is claimed optimal.

1 Model and specification

A circuit is an ordered list of 2-input XOR gates: $s_0, \dots, s_{31}$ are the input bits, and gate k produces $s_{32+k} = s_a \oplus s_b$ with $a, b < 32 + k$. Cost is the gate count; depth counts gates on the longest input-to-output path, inputs at depth 0. Bit index i denotes bit $(i \bmod 8)$ of state byte $\lfloor i/8 \rfloor$, least-significant-bit first, and MixColumns maps an input column $a_0, \dots, a_3$ to

$$\text{out}[c] = 2 \cdot a_c \oplus 3 \cdot a_{(c+1) \bmod 4} \oplus a_{(c+2) \bmod 4} \oplus a_{(c+3) \bmod 4},$$

in $\text{GF}(2^8)$ modulo $x^8 + x^4 + x^3 + x + 1$ (0x11b), per NIST FIPS 197-upd1, Eq. 5.6 [1]. The map is $\text{GF}(2)$ -linear, so agreement on the 32 unit-input vectors is a complete correctness check; counts and depths are relabeling-invariant, so no comparison below depends on convention.

The s-XOR (in-place) model is comparable: SSA-unrolling a k -instruction s-XOR program yields exactly k free-standing gates. Multi-input XOR gates, gate-equivalent area and quantum CNOT counts are different cost functions: not comparable, not claimed against.

2 Results

Depth 3 is the known minimum depth — an output depending on w inputs needs depth $\geq \lceil \log_2 w \rceil$, and MixColumns has outputs of weight 7 [2] — so the depth-3 circuit's contribution is its count. The 89 at depth 5 is shallower than any published circuit of fewer than 94 gates whose depth is stated, and stays on the frontier: the published sub-89 point, Jean's 88 [9], sits at depth 7, and [8] states none.

The 88 at depth 7 *ties* Jean's point rather than improving it: Jean's note was posted 2026-07-23 and has priority. Ours was found 2026-07-26 on a fully logged own lineage —

Depth	Published best	Model	This work	Relation
3	99 [2]	XOR2	97	−2 (depth 3 is the minimum depth)
4	97 [3]	XOR2	92	−5
5	94 [3]	XOR2	89	−5
6	92 [4, 5]	XOR2 / s-XOR	—	dominated by 89 @ 5
7	88 [9]	XOR2	88	ties it; independent, 61/88 masks shared
8	—	XOR2	(88) [†]	derived from [9]; dominated by 88 @ 7
n/s	89 [8]	g-XOR	—	depth not stated

Table 1: The published depth–count landscape in comparable models, and this work. [†]Derived and non-frontier: listed for completeness, claimed as nothing. Jean’s 88 [9] supersedes the earlier 91 of [6, 7] at depth 7; the depth-3 lineage is 105 → 103 → 102 → 99 ([2], Table 3). Listings: Appendices A–E.

from-scratch 97 at depth 3 (235 s), a 21 h ladder to 89 at depth 6 (11.0 h), one reroute to 89 at depth 5 (592 s), ρ^2 -symmetrization to a 94 at depth 5, then 32.9 min of plateau walking — with no imported circuit in the chain, and is a different circuit: 61 of 88 masks shared, Jaccard 0.530. The fifth circuit, 88 at depth 8 (Appendix E), is derived work and dominated: its seed chain passes through Jean’s, ρ^2 -symmetrized and peeled to 95, orbit-walked to 92, then unioned with a 91 of our own lineage before the descent.

3 Method

A circuit is represented as a *value set*: masks over the 32 inputs, valid iff every mask is the XOR of two already-available masks and all 32 output masks are present; gate count is the set size. Deletion becomes a set operation, circuits compare by mask overlap regardless of gate order, and a neighbourhood — remove k masks, add at most $k - 1$ — becomes finite and enumerable (Section 4).

In published circuits every gate is locally efficient, the signature of greedy construction; smaller circuits are separated from them by locally *suboptimal* steps. **Remove-1** drops a non-output mask if the set stays realizable. The **neutral swap** removes one mask and adds one repair mask at unchanged size: with $A = \text{closure}(S')$ and $stuck = (S' \cup T) \setminus A$, every single-mask repair lies in $C = \{u \oplus a : u \in stuck, a \in A\} \cap P_2(A)$ (P_2 = pair sums), so enumerating C *decides* the move — measured $|C| \approx 7$, cheaper than the earlier engine’s 24–40 random candidates, which missed 81–82 % of existing repairs. **Remove-2-add-1** remains a diversifier, though at the 88/89 frontier it is provably empty (Section 4). **Destroy-and-rebuild** removes a group and re-synthesizes from a pool, accepting rebuilds no larger, or larger on an annealing schedule with reheat; destroying a *connected cone* of the DAG leaves a hole worth re-planning, where k unrelated masks are usually just re-derived, and near-miss rebuilds are peeled before judgement. A fixpoint over minimum build depths rejects cap violations and emits gate lists in depth order.

Workers run one engine at one depth cap in fixed-length chunks, alternating walk and rebuild, and a coordinator re-offers each the best circuit feasible at its cap (the reseeding ladder). A worker accepts fewer gates *or equal gates at strictly lower depth* — a Pareto tie-break, and not a cosmetic one: 89 at depth 6 became 89 at depth 5 through it, and both 88s, first seen at depths 11 and 10, reached depths 7 and 8 seconds later. Every candidate is verified against MixColumns rebuilt from $\text{GF}(2^8)$ before acceptance. The depth-3 circuit comes from a separate constructor: each output is modelled as $A \oplus B$, A and B partitioning its bits into shared, refcount-costed depth- ≤ 2 parts, annealed by iterated local search, so depth ≤ 3 holds by construction.

The v2 engine. Version 1’s circuits came from an engine whose kernel was then profiled and rewritten [10]: level-BFS depth relaxation ($17.5\times$ on destroy-and-rebuild, bit-identical trajectories); a worklist closure with incremental removal queries, and removal of a bug that re-verified unchanged circuits every plateau iteration — 78 % of walk runtime ($21.9\times$ on the walk); victim repooling ($\sim 34\times$ accepted moves/s); cone destroys ($\sim 12\times$); exact repair ($\sim 7\times$ plateau mobility). The walk now runs at 480–640 iterations/s against 70–110. What found both 88s was not a cleverer move but a cheap plateau walked at volume: every distinct equal-best mask set is harvested.

4 Exact local certificates, and structure

To say what is *not* there we decide the neighbourhood question: given a verified 88-gate circuit and a window of k removed masks, can at most $k - 1$ new masks — each buildable when added, with cascade unlocks between — restore all 32 targets? A “yes” is an 87-gate circuit; a completed “no” is a proof for that window. Completeness follows from a first-unlock case analysis: the first unlock must have the newest added mask as a parent, which pins each added mask to a finite, enumerable shape. Runs above the proved budget are recorded as undecided, never as irreducible. The procedure agrees with an independent brute force 25/25 and 12/12 at $k \leq 3$, and on 1 257 instances including 122 genuine negatives at $k = 4$.

- **47 canonical 88-gate circuits** — Jean’s 88, its 12 plateau siblings, and a third family’s anchor with 33 representatives — **have exhaustively empty $k \leq 3$ shells**: all 1 540 windows at $k = 2$ and all 27 720 at $k = 3$ each, none reducible. Hence *any 87-gate circuit differs from each of the 47 by at least 4 masks*. Our own 88 at depth 7 is *not* among the 47: its exhaustive $k \leq 3$ sweep was never run; it has 9 exact $k = 4$ windows and 8 SAT cone windows instead.
- **Population and depth census.** Harvesting mapped $\approx 139\,878$ distinct 88-gate mask sets, of which 105 801 are proven irreducible at $k = 2$. Every one has minimum build depth ≥ 7 , and 18 353 (13.1 %) reach depth exactly 7: no 88-gate circuit of depth ≤ 6 has been seen (depths recomputed over the whole harvest, itself listed by size and SHA-256 in [10]). In total ≈ 165 million exact window decisions, zero timeouts, zero reducible windows; all five 89-gate seeds likewise have empty $k = 2$ shells.
- **Windowed SAT is evidence, not proof.** A sound windowed CNF encoding, solved by Kissat under two soundness-proved symmetry-breaking layers, returned UNSAT up to $k = 16$ on Jean’s 88 and $k = 15$ on the third-family anchor (whose whole $k \leq 12$ regime is UNSAT, 28/28), with **0 SAT anywhere** — but UNSAT there is relative to the encoding’s fixed slot order, and undecided windows are undecided.

None of this bounds 87 away globally; it is consistent with 88 being locally rigid, and it redirected the search: at 88 gates every move class small enough to enumerate is empty, so what remains is walking the equal-size plateau widely.

Symmetry. Byte rotation ρ (order 4) commutes with MixColumns, so the 32 targets form 8 ρ -orbits of size 4 and equivariant circuits can be searched in orbit space — but full ρ -symmetry costs about +19 gates. The half-rotation ρ^2 is where the structure is: in the elite basin 79–81 % of a circuit’s masks have their ρ^2 -image also in the mask set, and the 12 ρ^2 -fixed masks the search settles on are exactly the classic $x_0 \oplus x_2 / x_1 \oplus x_3$ pairs, rediscovered rather than imposed. Jean’s 88, at 75 % (66 of 88), symmetrizes into a second, distinct basin. A ρ^2 -equivariant orbit engine produced two *exactly* ρ^2 -symmetric 90-gate circuits (depths 9 and

7), improving the best exactly symmetric circuit we know of from 94 to 90; both are certified locally optimal in orbit space, and both are derived work, each seed union containing a circuit symmetrized from Jean’s 88. Symmetry is also a seeding device: the exactly ρ^2 -symmetric 94 at depth 5 — 82 of its 94 masks shared with the 89 at depth 5 — costs +5 over that 89 and is the seed from which the 88 at depth 7 was found.

5 Verification, artifacts, and reproduction

Every circuit here is checkable from the note alone: the appendices list every gate, Section 1 defines the target map, and because MixColumns is linear, replaying a listing on the 32 unit-input vectors against Eq. 5.6 [1] is a complete correctness check. Each appendix also states its circuit’s canonical SHA-256 over the UTF-8 bytes of `{"inputCount":32,"gates":[...]}`.

The artifacts automate that. Circuits and verifiers are at <https://github.com/Joe-b-20/aes-mixcolumns-xor-circuits>, archived at concept DOI 10.5281/zenodo.21299092; `python3 verify_all.py` runs `verify.py`, which rebuilds the target map from the $\text{GF}(2^8)$ definition so the verifier *is* the specification, a clean-room verifier with adversarial rejection tests, and Verilog testbenches, all under continuous integration. The method, pipeline, untouched run archives with each run’s exact code, certificate summaries and the credited transcriptions of the imported circuits [9, 8] are published separately [10], one reproduction command per result. Measured there on 2026-07-27 with the shipped v2 engine, and reported as measurements rather than promises: the from-scratch 97 at depth 3 in 81 s on one core; the 89 at depth 5 in 19 s and 22 s from shipped seeds, against 592 s for the archived 2026-07-14 run on the v1 engine; the 88 at depth 7 in 19.4 and 31.0 minutes in two single-worker re-runs from the ρ^2 -symmetric 94, against 32.9 min in the archived ten-worker run; the 92 at depth 4 a from-scratch cascade of hours. The re-runs are re-runs, not independent confirmations.

Implementation and execution used LLM coding agents as programming tools, directed by the author, who designed the moves and acceptance rules. The published method is fully specified and reproducible with the accompanying code, with no AI system in the loop.

6 Scope and claims

- **Not optimality.** Minimum 2-input-XOR circuit size (the Shortest Linear Program problem) is NP-hard; no count here is proven minimal, and the certificates of Section 4 are local.
- **Comparisons.** All counts are for the model and convention of Section 1; the baselines [2, 3, 4, 5, 6, 7, 8, 9] were source-checked, and a source-by-source audit, with a dated Corrections section, is in the artifact repository’s `PRIOR_ART.md`. We claim the smallest counts at depths 3, 4 and 5, and no improvement at unconstrained depth: the published floor is 88 and stays 88.
- **Priority.** The 88 at depth 7 ties [9], posted first; the 88 at depth 8 is derived from that circuit and dominated. Neither is a record claim. The depths quoted for [9] and [8] — 7 and 9 — are our measurements of our own transcriptions; neither paper states a depth.
- **Earlier results.** This note supersedes the project’s 98 at depth 3, 91 at depth 6 and 89 at depth 10, kept in the artifact repository.

Version history. **v2**, 27 July 2026: adds the 88 at depth 7 and its derived depth-8 companion, the exact neighbourhood certificates, the depth census and ρ^2 symmetry results, the

v2 engine measurements and dated reproduction times; no claim of v1 is withdrawn. **v1**, July 2026, archived at version DOI 10.5281/zenodo.21299093: the 97, 92 and 89 only.

References

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- [10] J. Bachir, *The method: value-set SLP search, plateau walking, and exact local certificates* (METHODS.md, evidence/RESULTS.md, reproduce/), slp-plateau-search v2.0.0, 2026. <https://github.com/Joe-b-20/slp-plateau-search>

A Circuit mixcolumns_97gates_depth3 (97 gates, depth 3)

Signals $s_0, \dots, s_{31}$ are the inputs; gate k produces s_{32+k} . Canonical gate-list SHA-256 (see Section 5):

e4c4e678c4b7c22d889c6e060b43859c111b82b953d6d9b43c49c079c0545b56

$s_{32} = s_0 \oplus s_8$	$s_{40} = s_{11} \oplus s_{20}$	$s_{48} = s_{11} \oplus s_{26}$	$s_{56} = s_{22} \oplus s_{30}$
$s_{33} = s_5 \oplus s_{13}$	$s_{41} = s_{12} \oplus s_{20}$	$s_{49} = s_2 \oplus s_{27}$	$s_{57} = s_7 \oplus s_{31}$
$s_{34} = s_6 \oplus s_{14}$	$s_{42} = s_6 \oplus s_{22}$	$s_{50} = s_3 \oplus s_{27}$	$s_{58} = s_{23} \oplus s_{31}$
$s_{35} = s_7 \oplus s_{15}$	$s_{43} = s_{15} \oplus s_{23}$	$s_{51} = s_4 \oplus s_{27}$	$s_{59} = s_{33} \oplus s_{14}$
$s_{36} = s_8 \oplus s_{16}$	$s_{44} = s_1 \oplus s_{24}$	$s_{52} = s_4 \oplus s_{28}$	$s_{60} = s_{35} \oplus s_{17}$
$s_{37} = s_9 \oplus s_{17}$	$s_{45} = s_1 \oplus s_{25}$	$s_{53} = s_{12} \oplus s_{28}$	$s_{61} = s_{39} \oplus s_{10}$
$s_{38} = s_{10} \oplus s_{18}$	$s_{46} = s_9 \oplus s_{25}$	$s_{54} = s_{13} \oplus s_{29}$	$s_{62} = s_{41} \oplus s_5$
$s_{39} = s_{11} \oplus s_{19}$	$s_{47} = s_2 \oplus s_{26}$	$s_{55} = s_{21} \oplus s_{29}$	$s_{63} = s_{35} \oplus s_{40}$

s64 = s42 $\oplus$ s7	s81 = s51 $\oplus$ s19	s98 = s70 $\oplus$ s60	s115 = s75 $\oplus$ s71
s65 = s36 $\oplus$ s43	s82 = s51 $\oplus$ s41	s99 = s74 $\oplus$ s77	s116 = s95 $\oplus$ s79
s66 = s43 $\oplus$ s19	s83 = s53 $\oplus$ s3	s100 = s78 $\oplus$ s61	s117 = s96 $\oplus$ s81
s67 = s32 $\oplus$ s24	s84 = s52 $\oplus$ s40	s101 = s63 $\oplus$ s83	s118 = s87 $\oplus$ s62
s68 = s36 $\oplus$ s24	s85 = s52 $\oplus$ s21	s102 = s87 $\oplus$ s85	s119 = s89 $\oplus$ s34
s69 = s44 $\oplus$ s37	s86 = s42 $\oplus$ s54	s103 = s59 $\oplus$ s56	s120 = s92 $\oplus$ s56
s70 = s32 $\oplus$ s46	s87 = s53 $\oplus$ s54	s104 = s93 $\oplus$ s15	s121 = s90 $\oplus$ s36
s71 = s46 $\oplus$ s10	s88 = s43 $\oplus$ s56	s105 = s65 $\oplus$ s67	s122 = s69 $\oplus$ s90
s72 = s46 $\oplus$ s16	s89 = s55 $\oplus$ s30	s106 = s65 $\oplus$ s73	s123 = s74 $\oplus$ s2
s73 = s45 $\oplus$ s17	s90 = s57 $\oplus$ s0	s107 = s75 $\oplus$ s18	s124 = s76 $\oplus$ s91
s74 = s45 $\oplus$ s38	s91 = s57 $\oplus$ s3	s108 = s80 $\oplus$ s66	s125 = s82 $\oplus$ s91
s75 = s47 $\oplus$ s37	s92 = s57 $\oplus$ s15	s109 = s84 $\oplus$ s66	s126 = s85 $\oplus$ s33
s76 = s47 $\oplus$ s39	s93 = s34 $\oplus$ s58	s110 = s62 $\oplus$ s55	s127 = s86 $\oplus$ s59
s77 = s46 $\oplus$ s26	s94 = s44 $\oplus$ s58	s111 = s86 $\oplus$ s89	s128 = s88 $\oplus$ s64
s78 = s49 $\oplus$ s35	s95 = s48 $\oplus$ s58	s112 = s93 $\oplus$ s64	
s79 = s50 $\oplus$ s18	s96 = s53 $\oplus$ s58	s113 = s67 $\oplus$ s58	
s80 = s50 $\oplus$ s38	s97 = s68 $\oplus$ s35	s114 = s94 $\oplus$ s72	

Output signals (output bit j is carried by the listed signal, $j = 0, \dots, 31$):

[97, 98, 99, 100, 101, 102, 103, 104, 105, 106, 107, 108, 109, 110, 111, 112, 113, 114, 115, 116, 117, 118, 119, 120, 121, 122, 123, 124, 125, 126, 127, 128]

B Circuit mixcolumns_92gates_depth4 (92 gates, depth 4)

Signals $s_0, \dots, s_{31}$ are the inputs; gate k produces s_{32+k} . Canonical gate-list SHA-256 (see Section 5):

e79bfccf5322fe8933242b47b53e94e73163615c70d5c45c9721e27ac86fd97d

s32 = s10 $\oplus$ s26	s56 = s15 $\oplus$ s31	s80 = s45 $\oplus$ s49	s104 = s63 $\oplus$ s49
s33 = s12 $\oplus$ s20	s57 = s43 $\oplus$ s44	s81 = s21 $\oplus$ s33	s105 = s64 $\oplus$ s50
s34 = s14 $\oplus$ s30	s58 = s34 $\oplus$ s53	s82 = s67 $\oplus$ s81	s106 = s64 $\oplus$ s69
s35 = s6 $\oplus$ s14	s59 = s5 $\oplus$ s54	s83 = s57 $\oplus$ s66	s107 = s23 $\oplus$ s75
s36 = s11 $\oplus$ s28	s60 = s0 $\oplus$ s37	s84 = s58 $\oplus$ s77	s108 = s71 $\oplus$ s80
s37 = s8 $\oplus$ s16	s61 = s38 $\oplus$ s52	s85 = s42 $\oplus$ s81	s109 = s46 $\oplus$ s72
s38 = s7 $\oplus$ s31	s62 = s6 $\oplus$ s40	s86 = s59 $\oplus$ s67	s110 = s102 $\oplus$ s66
s39 = s17 $\oplus$ s25	s63 = s32 $\oplus$ s56	s87 = s59 $\oplus$ s40	s111 = s18 $\oplus$ s101
s40 = s13 $\oplus$ s21	s64 = s36 $\oplus$ s56	s88 = s10 $\oplus$ s73	s112 = s89 $\oplus$ s78
s41 = s2 $\oplus$ s26	s65 = s37 $\oplus$ s56	s89 = s45 $\oplus$ s79	s113 = s63 $\oplus$ s108
s42 = s5 $\oplus$ s29	s66 = s1 $\oplus$ s48	s90 = s23 $\oplus$ s68	s114 = s108 $\oplus$ s55
s43 = s16 $\oplus$ s24	s67 = s50 $\oplus$ s53	s91 = s62 $\oplus$ s52	s115 = s93 $\oplus$ s65
s44 = s9 $\oplus$ s25	s68 = s35 $\oplus$ s56	s92 = s58 $\oplus$ s62	s116 = s47 $\oplus$ s109
s45 = s18 $\oplus$ s26	s69 = s3 $\oplus$ s38	s93 = s60 $\oplus$ s76	s117 = s52 $\oplus$ s107
s46 = s0 $\oplus$ s17	s70 = s51 $\oplus$ s54	s94 = s35 $\oplus$ s77	s118 = s96 $\oplus$ s104
s47 = s1 $\oplus$ s9	s71 = s19 $\oplus$ s48	s95 = s15 $\oplus$ s61	s119 = s33 $\oplus$ s106
s48 = s23 $\oplus$ s31	s72 = s24 $\oplus$ s38	s96 = s69 $\oplus$ s74	s120 = s71 $\oplus$ s105
s49 = s3 $\oplus$ s27	s73 = s39 $\oplus$ s41	s97 = s65 $\oplus$ s72	s121 = s103 $\oplus$ s51
s50 = s4 $\oplus$ s20	s74 = s41 $\oplus$ s55	s98 = s38 $\oplus$ s60	s122 = s97 $\oplus$ s100
s51 = s12 $\oplus$ s27	s75 = s34 $\oplus$ s38	s99 = s70 $\oplus$ s71	s123 = s90 $\oplus$ s75
s52 = s22 $\oplus$ s30	s76 = s43 $\oplus$ s48	s100 = s57 $\oplus$ s46	
s53 = s13 $\oplus$ s29	s77 = s22 $\oplus$ s42	s101 = s39 $\oplus$ s78	
s54 = s4 $\oplus$ s28	s78 = s41 $\oplus$ s44	s102 = s39 $\oplus$ s65	
s55 = s11 $\oplus$ s19	s79 = s10 $\oplus$ s47	s103 = s69 $\oplus$ s50	

Output signals (output bit j is carried by the listed signal, $j = 0, \dots, 31$):

[97, 122, 89, 118, 119, 82, 84, 90, 115, 110, 111, 113, 120, 85, 91, 117, 93, 83, 88, 114, 99, 86, 92, 95, 98, 116, 112, 96, 121, 87, 94, 123]

C Circuit mixcolumns_89gates_depth5 (89 gates, depth 5)

Signals $s_0, \dots, s_{31}$ are the inputs; gate k produces s_{32+k} . Canonical gate-list SHA-256 (see Section 5):

c3b972af282d25e8313a510f5cf840472401c6268bd28cda889593fc3379f064

s32 = s9 $\oplus$ s17	s55 = s21 $\oplus$ s39	s78 = s55 $\oplus$ s66	s101 = s60 $\oplus$ s50
s33 = s19 $\oplus$ s27	s56 = s5 $\oplus$ s46	s79 = s37 $\oplus$ s57	s102 = s39 $\oplus$ s101
s34 = s1 $\oplus$ s25	s57 = s14 $\oplus$ s42	s80 = s35 $\oplus$ s56	s103 = s34 $\oplus$ s87
s35 = s21 $\oplus$ s29	s58 = s15 $\oplus$ s37	s81 = s56 $\oplus$ s66	s104 = s59 $\oplus$ s89
s36 = s12 $\oplus$ s28	s59 = s11 $\oplus$ s33	s82 = s55 $\oplus$ s42	s105 = s76 $\oplus$ s70
s37 = s22 $\oplus$ s30	s60 = s11 $\oplus$ s43	s83 = s70 $\oplus$ s72	s106 = s92 $\oplus$ s65
s38 = s3 $\oplus$ s27	s61 = s0 $\oplus$ s52	s84 = s43 $\oplus$ s71	s107 = s96 $\oplus$ s98
s39 = s4 $\oplus$ s28	s62 = s33 $\oplus$ s45	s85 = s54 $\oplus$ s67	s108 = s36 $\oplus$ s93
s40 = s9 $\oplus$ s25	s63 = s38 $\oplus$ s52	s86 = s54 $\oplus$ s49	s109 = s95 $\oplus$ s48
s41 = s13 $\oplus$ s29	s64 = s43 $\oplus$ s51	s87 = s17 $\oplus$ s64	s110 = s1 $\oplus$ s92
s42 = s5 $\oplus$ s13	s65 = s8 $\oplus$ s45	s88 = s57 $\oplus$ s67	s111 = s99 $\oplus$ s46
s43 = s15 $\oplus$ s23	s66 = s36 $\oplus$ s41	s89 = s60 $\oplus$ s47	s112 = s98 $\oplus$ s75
s44 = s6 $\oplus$ s22	s67 = s41 $\oplus$ s44	s90 = s62 $\oplus$ s53	s113 = s89 $\oplus$ s90
s45 = s7 $\oplus$ s23	s68 = s34 $\oplus$ s53	s91 = s71 $\oplus$ s74	s114 = s108 $\oplus$ s99
s46 = s12 $\oplus$ s20	s69 = s24 $\oplus$ s45	s92 = s24 $\oplus$ s61	s115 = s110 $\oplus$ s112
s47 = s10 $\oplus$ s18	s70 = s40 $\oplus$ s47	s93 = s38 $\oplus$ s62	s116 = s3 $\oplus$ s104
s48 = s2 $\oplus$ s26	s71 = s31 $\oplus$ s49	s94 = s58 $\oplus$ s52	s117 = s90 $\oplus$ s109
s49 = s6 $\oplus$ s14	s72 = s26 $\oplus$ s34	s95 = s59 $\oplus$ s63	s118 = s108 $\oplus$ s101
s50 = s19 $\oplus$ s20	s73 = s32 $\oplus$ s48	s96 = s64 $\oplus$ s69	s119 = s87 $\oplus$ s112
s51 = s8 $\oplus$ s16	s74 = s44 $\oplus$ s45	s97 = s61 $\oplus$ s51	s120 = s32 $\oplus$ s110
s52 = s7 $\oplus$ s31	s75 = s40 $\oplus$ s51	s98 = s0 $\oplus$ s65	
s53 = s2 $\oplus$ s18	s76 = s18 $\oplus$ s73	s99 = s4 $\oplus$ s63	
s54 = s30 $\oplus$ s35	s77 = s10 $\oplus$ s68	s100 = s58 $\oplus$ s74	

Output signals (output bit j is carried by the listed signal, $j = 0, \dots, 31$):

[96, 119, 83, 113, 118, 78, 79, 84, 107, 103, 76, 116, 102, 80, 85, 91, 106, 115, 105, 117, 114, 81, 86, 94, 97, 120, 77, 109, 111, 82, 88, 100]

D Circuit mixcolumns_88gates_depth7 (88 gates, depth 7; ties the published floor)

Status: the same (depth, count) point as Jean [9], reached independently on this project's own lineage; 61 of 88 internal masks are shared. It ties that point and does not beat it, and Jean has priority.

Signals $s_0, \dots, s_{31}$ are the inputs; gate k produces s_{32+k} . Canonical gate-list SHA-256 (see Section 5):

76bac50ab0e4a7dd54578e77f57f6d08e96a8803aff72dba2c3f5fadaaa06ee1

s32 = s13 $\oplus$ s29	s55 = s34 $\oplus$ s15	s78 = s52 $\oplus$ s64	s101 = s87 $\oplus$ s37
s33 = s21 $\oplus$ s29	s56 = s35 $\oplus$ s20	s79 = s53 $\oplus$ s63	s102 = s90 $\oplus$ s61
s34 = s22 $\oplus$ s30	s57 = s35 $\oplus$ s8	s80 = s53 $\oplus$ s59	s103 = s90 $\oplus$ s40
s35 = s15 $\oplus$ s23	s58 = s36 $\oplus$ s2	s81 = s55 $\oplus$ s68	s104 = s94 $\oplus$ s38
s36 = s17 $\oplus$ s25	s59 = s37 $\oplus$ s5	s82 = s57 $\oplus$ s16	s105 = s94 $\oplus$ s40
s37 = s12 $\oplus$ s20	s60 = s39 $\oplus$ s1	s83 = s58 $\oplus$ s61	s106 = s95 $\oplus$ s60
s38 = s12 $\oplus$ s28	s61 = s39 $\oplus$ s51	s84 = s59 $\oplus$ s33	s107 = s96 $\oplus$ s44
s39 = s9 $\oplus$ s25	s62 = s41 $\oplus$ s50	s85 = s60 $\oplus$ s17	s108 = s96 $\oplus$ s36
s40 = s2 $\oplus$ s18	s63 = s42 $\oplus$ s21	s86 = s62 $\oplus$ s76	s109 = s97 $\oplus$ s108
s41 = s3 $\oplus$ s27	s64 = s43 $\oplus$ s14	s87 = s62 $\oplus$ s4	s110 = s98 $\oplus$ s49
s42 = s4 $\oplus$ s28	s65 = s43 $\oplus$ s63	s88 = s64 $\oplus$ s34	s111 = s100 $\oplus$ s105
s43 = s5 $\oplus$ s13	s66 = s44 $\oplus$ s45	s89 = s66 $\oplus$ s86	s112 = s104 $\oplus$ s87
s44 = s15 $\oplus$ s31	s67 = s45 $\oplus$ s58	s90 = s67 $\oplus$ s85	s113 = s104 $\oplus$ s71
s45 = s10 $\oplus$ s26	s68 = s46 $\oplus$ s49	s91 = s68 $\oplus$ s70	s114 = s106 $\oplus$ s108
s46 = s6 $\oplus$ s22	s69 = s47 $\oplus$ s54	s92 = s70 $\oplus$ s35	s115 = s107 $\oplus$ s74
s47 = s6 $\oplus$ s14	s70 = s47 $\oplus$ s31	s93 = s71 $\oplus$ s42	s116 = s107 $\oplus$ s110
s48 = s11 $\oplus$ s19	s71 = s48 $\oplus$ s56	s94 = s72 $\oplus$ s3	s117 = s107 $\oplus$ s57
s49 = s7 $\oplus$ s23	s72 = s49 $\oplus$ s19	s95 = s73 $\oplus$ s74	s118 = s111 $\oplus$ s41
s50 = s7 $\oplus$ s31	s73 = s49 $\oplus$ s16	s96 = s73 $\oplus$ s0	s119 = s111 $\oplus$ s66
s51 = s18 $\oplus$ s26	s74 = s50 $\oplus$ s24	s97 = s82 $\oplus$ s9	
s52 = s32 $\oplus$ s46	s75 = s50 $\oplus$ s55	s98 = s82 $\oplus$ s24	
s53 = s32 $\oplus$ s38	s76 = s51 $\oplus$ s72	s99 = s85 $\oplus$ s97	
s54 = s33 $\oplus$ s30	s77 = s52 $\oplus$ s54	s100 = s86 $\oplus$ s48	

Output signals (output bit j is carried by the listed signal, $j = 0, \dots, 31$):

[110, 109, 103, 119, 113, 79, 88, 92, 115, 99, 83, 89, 93, 84, 77, 91, 116, 106, 67, 100, 112, 80, 69, 75, 117, 114, 102, 118, 101, 65, 78, 81]

E Circuit mixcolumns_88gates_depth8 (88 gates, depth 8; derived, non-frontier)

Status: derived from published work, and not a frontier point. Its seed chain passes through Jean's published 88-gate circuit [9] (Section 2), and it is dominated by Appendix D (same count, greater depth). It is listed as a verified distinct construction, not as a claim.

Signals $s_0, \dots, s_{31}$ are the inputs; gate k produces s_{32+k} . Canonical gate-list SHA-256 (see Section 5):

3a23e2f2b2f3835cab74ba23800088bb03ab4ee3f076f2b9a826c78cd84f8ef1

$s_{32} = s_{13} \oplus s_{29}$	$s_{55} = s_{35} \oplus s_8$	$s_{78} = s_{59} \oplus s_{75}$	$s_{101} = s_{95} \oplus s_{20}$
$s_{33} = s_{21} \oplus s_{29}$	$s_{56} = s_{38} \oplus s_{26}$	$s_{79} = s_{59} \oplus s_{63}$	$s_{102} = s_{96} \oplus s_{16}$
$s_{34} = s_{22} \oplus s_{30}$	$s_{57} = s_{39} \oplus s_{50}$	$s_{80} = s_{60} \oplus s_{34}$	$s_{103} = s_{96} \oplus s_8$
$s_{35} = s_{15} \oplus s_{23}$	$s_{58} = s_{40} \oplus s_{21}$	$s_{81} = s_{61} \oplus s_{41}$	$s_{104} = s_{97} \oplus s_{36}$
$s_{36} = s_9 \oplus s_{17}$	$s_{59} = s_{42} \oplus s_{45}$	$s_{82} = s_{63} \oplus s_{76}$	$s_{105} = s_{100} \oplus s_{89}$
$s_{37} = s_{12} \oplus s_{28}$	$s_{60} = s_{43} \oplus s_{14}$	$s_{83} = s_{64} \oplus s_{66}$	$s_{106} = s_{100} \oplus s_{39}$
$s_{38} = s_2 \oplus s_{18}$	$s_{61} = s_{43} \oplus s_{58}$	$s_{84} = s_{66} \oplus s_{35}$	$s_{107} = s_{101} \oplus s_{94}$
$s_{39} = s_3 \oplus s_{27}$	$s_{62} = s_{45} \oplus s_{19}$	$s_{85} = s_{67} \oplus s_{19}$	$s_{108} = s_{101} \oplus s_{37}$
$s_{40} = s_4 \oplus s_{28}$	$s_{63} = s_{46} \oplus s_{26}$	$s_{86} = s_{67} \oplus s_{11}$	$s_{109} = s_{102} \oplus s_{44}$
$s_{41} = s_4 \oplus s_{20}$	$s_{64} = s_{47} \oplus s_{49}$	$s_{87} = s_{68} \oplus s_0$	$s_{110} = s_{102} \oplus s_{68}$
$s_{42} = s_9 \oplus s_{25}$	$s_{65} = s_{48} \oplus s_{53}$	$s_{88} = s_{74} \oplus s_{17}$	$s_{111} = s_{106} \oplus s_{86}$
$s_{43} = s_5 \oplus s_{13}$	$s_{66} = s_{48} \oplus s_{31}$	$s_{89} = s_{76} \oplus s_{44}$	$s_{112} = s_{106} \oplus s_{38}$
$s_{44} = s_{15} \oplus s_{31}$	$s_{67} = s_{49} \oplus s_3$	$s_{90} = s_{77} \oplus s_{41}$	$s_{113} = s_{109} \oplus s_{68}$
$s_{45} = s_{10} \oplus s_{18}$	$s_{68} = s_{50} \oplus s_{24}$	$s_{91} = s_{81} \oplus s_{33}$	$s_{114} = s_{110} \oplus s_{42}$
$s_{46} = s_1 \oplus s_{25}$	$s_{69} = s_{50} \oplus s_{54}$	$s_{92} = s_{81} \oplus s_{51}$	$s_{115} = s_{112} \oplus s_{93}$
$s_{47} = s_6 \oplus s_{22}$	$s_{70} = s_{51} \oplus s_{58}$	$s_{93} = s_{85} \oplus s_{89}$	$s_{116} = s_{113} \oplus s_{103}$
$s_{48} = s_6 \oplus s_{14}$	$s_{71} = s_{52} \oplus s_{53}$	$s_{94} = s_{85} \oplus s_{40}$	$s_{117} = s_{113} \oplus s_{55}$
$s_{49} = s_7 \oplus s_{23}$	$s_{72} = s_{52} \oplus s_{60}$	$s_{95} = s_{86} \oplus s_{35}$	$s_{118} = s_{114} \oplus s_{97}$
$s_{50} = s_7 \oplus s_{31}$	$s_{73} = s_{54} \oplus s_{64}$	$s_{96} = s_{87} \oplus s_{49}$	$s_{119} = s_{114} \oplus s_{88}$
$s_{51} = s_{32} \oplus s_{37}$	$s_{74} = s_{55} \oplus s_{16}$	$s_{97} = s_{87} \oplus s_1$	
$s_{52} = s_{32} \oplus s_{47}$	$s_{75} = s_{56} \oplus s_{36}$	$s_{98} = s_{88} \oplus s_{46}$	
$s_{53} = s_{33} \oplus s_{30}$	$s_{76} = s_{56} \oplus s_{10}$	$s_{99} = s_{94} \oplus s_{77}$	
$s_{54} = s_{34} \oplus s_{15}$	$s_{77} = s_{57} \oplus s_{12}$	$s_{100} = s_{95} \oplus s_{62}$	

Output signals (output bit j is carried by the listed signal, $j = 0, \dots, 31$):

[116, 119, 79, 112, 108, 70, 80, 84, 109, 98, 75, 111, 107, 92, 71, 83, 103, 118, 78, 115, 99, 91, 65, 69, 117, 104, 82, 105, 90, 61, 72, 73]